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qwen35-toolkit

Python toolkit for converting, stripping, verifying and publishing Qwen3.5 models to HuggingFace Hub.

> **Scope:** This repo covers model preparation — BNB quantization, visual tower removal, sanity verification, and Hub sync. LoRA training and post-training merging are out of scope and live in a separate repo.

Published Models

All models are available on HuggingFace Hub in four collections:

Collection	Description	Models
Qwen3.5 BNB 4-bit	VLM, BNB NF4 quantized	0.8B · 2B · 4B · 9B
Qwen3.5 Text	Text-only bf16	0.8B · 2B · 4B · 9B
Qwen3.5 Text BNB 4-bit	Text-only, BNB NF4	0.8B · 2B · 4B · 9B
Qwen3.5 Text GGUF	Text-only, GGUF Q4 - F16	0.8B · 2B · 4B · 9B

Setup

System

```
# Python 3.11 (required — bitsandbytes wheels target 3.11)
# I'm using Arch btw
yay -S python311

# Create and activate virtualenv
python3.11 -m venv venv
source venv/bin/activate
```

Installation

```
# 1. Install PyTorch with CUDA support first
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu124

# 2. Install the toolkit
pip install -e .
```

llama.cpp

```
git clone https://github.com/ggml-org/llama.cpp
```

```
# Build with CUDA support
cd llama.cpp
cmake -B build -DLLAMA_CUDA=ON
cmake --build build --config Release -j$(nproc)

# Running GGUF models with llama.cpp
# The GGUF models were quantized to Q4_K_M and are available at:
# https://huggingface.co/collections/techwithsergiu/qwen35-text-gguf
./build/bin/llama-cli \
  --hf-repo techwithsergiu/Qwen3.5-text-0.8B-GGUF \
  --hf-file Qwen3.5-text-0.8B-Q4_K_M.gguf \
  -p "Write a Python function that reverses a string." \
  -n 512

cd ..
```

Node.js (for PDF export only)

```
npm install -g @mermaid-js/mermaid-cli
npx puppeteer browsers install chrome-headless-shell
```

Commands

CLI command	Module	Purpose	Key args
`qwen35-convert`	`qwen35_toolkit.convert`	f16 → BNB nf4 ⚠️ Qwen3.5 only	`--model` `--output` `--low-vram` `--verbose` `--hf-token`
`qwen35-strip`	`qwen35_toolkit.strip`	Remove visual tower → text-only	`--model` `--output` `--mode` `--hf-token`

CLI command	Module	Purpose	Key args
`qwen35-verify`	`qwen35_toolkit.verify`	Universal sanity-check — BNB 4-bit **and** f16/bf16, auto-detected	`--model` `--hf-token`
`qwen35-verify-qwen35`	`qwen35_toolkit.verify_qwen35`	Qwen3.5 check — thinking ON/OFF + image	`--model` `--hf-token`
`qwen35-upload`	`qwen35_toolkit.upload`	Sync with HF Hub — push and pull	`--local` `--repo` `--mode` `--files` `--message` `--hf-token` `--private`
`qwen35-inspect`	`qwen35_toolkit.tools.inspect_safetensors`	Print weight prefixes and sizes from safetensors headers	positional: model path
`qwen35-render-mermaid`	`qwen35_toolkit.tools.render_mermaid`	Pre-render Mermaid diagrams → PNG for PDF export	`--input` `--output` `--scale`
—	—	GGUF f16 conversion (llama.cpp, external)	positional: model path, `--outtype`,

CLI command	Module	Purpose	Key args
			`--outfile`
—	—	GGUF quantization (llama.cpp, external)	positional: input, output, type

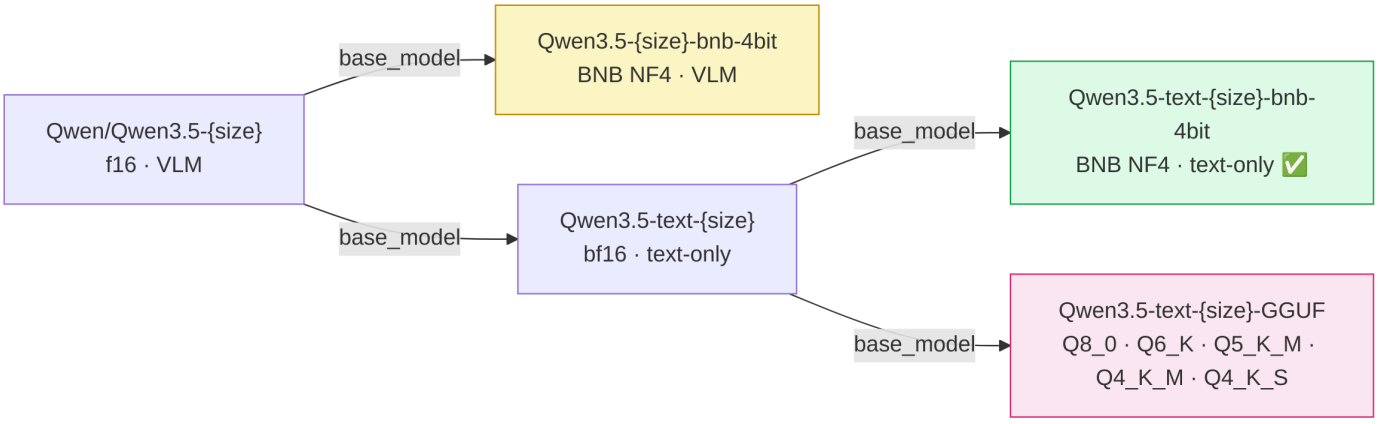
All commands support `--help` for full usage.

Hardware

GPU	RTX 3070 Laptop, 7.7 GB VRAM
BNB quantization (4B, default)	~5 GB VRAM
BNB quantization (any size, --low-vram)	~400 MB VRAM peak
f16 model load for strip/verify	~18 GB RAM
CPU merge	~20 GB RAM (not VRAM)
GGUF conversion (f16 → GGUF)	~18 GB RAM, no VRAM
GGUF Q4_K_M inference (4B)	~5 GB VRAM

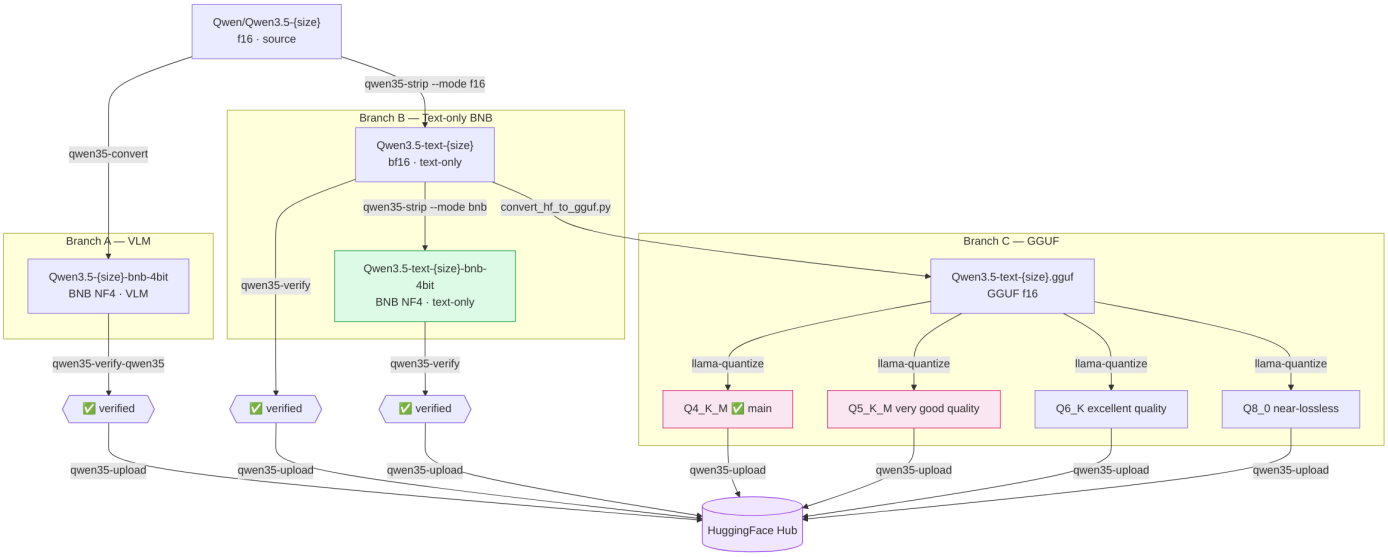
Model lineage

Four model families produced from the original Qwen3.5 f16. Arrows show the `base_model` relationship declared in each HF repo.



> **Why GGUF `base_model` points to `Qwen3.5-text-{size}`:** The GGUF quants are derived from the text-only f16 model — same weights, different container format. Linking to the f16 text variant keeps the lineage clear and separates GGUF from the VLM branch on the Hub.

Pipeline diagram



HuggingFace README templates

Four template files — one per model family. Replace `{SIZE}` with the model size (e.g. `0.8B`, `2B`, `4B`, `9B`).

File	Model type	`pipeline_tag`	`base_model`
`model_cards/README_Qwen3.5-bnb-4bit.md`	BNB 4-bit VLM	`image-text-to-text`	`Qwen/Qwen3.5-{SIZE}`
`model_cards/README_Qwen3.5-text.md`	f16 text-only	`text-generation`	`Qwen/Qwen3.5-{SIZE}`
`model_cards/README_Qwen3.5-text-bnb-4bit.md`	BNB 4-bit text-only	`text-generation`	`techwithsergiu/Qwen3.5-text-{SIZE}`

File	Model type	pipeline_tag	base_model
model_cards/README_Qwen3.5-text-GGUF.md	GGUF quants	text-generation	techwithsergiu/Qwen3.5-text-{SIZE}

Command reference

qwen35-convert

⚠️ Qwen3.5-specific — uses `Qwen3_5ForConditionalGeneration`.

Conversion strategy (avoids the `Params4bit/_is_hf_initialized` version mismatch bug):

```
1. Load full model into CPU RAM (bf16, no GPU involved)
2. Replace nn.Linear → Linear4bit in-place (still on CPU)
3. Quantize — BNB nf4 compression triggered by first .cuda() per weight.
   Two modes:
     default    : model.cuda() — full model to GPU at once.
     --low-vram : one layer at a time, each returned to CPU after quantization.
                  Peak VRAM ≈ one layer (~400 MB). Model saved from CPU.
4. Save quantized model + tokenizer
5. Rebuild config.json from source (architectures, vision_config,
quantization_config)
   Patch tokenizer_config.json (chat_template)
6. Copy optional side-car files (merges.txt, vocab.json) — missing files silently
skipped
```

Hardware requirements: RAM : ~18 GB (full bf16 model in RAM before quantization) VRAM : ~5 GB (default) or ~400 MB (`--low-vram`)

Layers skipped from quantization (left at full precision): `embed_tokens`, `embedding`, `lm_head`, `multi_modal_projector`, `merger`, `modality_projection`, `router`, `visual`, `vision_tower`.

The `quantization_config` written to `config.json` uses exact model-tree paths so transformers respects them on reload:

```
"llm_int8_skip_modules": [
    "lm_head",
    "model.visual",
    "model.language_model.embed_tokens"
]
```

Short names (`"visual"` etc.) are used only during the in-process conversion walk — they are not written to config.

`--verbose` prints every skip layer decision:

```
⏳ Replacing linear layers with BNB Linear4bit …
   └── skipped      model.visual
   └── skipped      model.language_model.embed_tokens
   └── skipped      lm_head
   186 layers replaced
```

Useful to confirm visual layers are correctly skipped (not quantized).

---

qwen35-strip

Removes the visual tower at the **file level** — no model load during stripping, operates directly on safetensors shards.

Two modes:

`--mode`	Input	Output	Use for
`bnb` (default)	BNB 4-bit VLM	text-only BNB 4-bit	training target
`f16`	f16 / bf16 VLM	text-only bf16	GGUF / merge base

```
1. Strip visual weights   — drop model.visual.*, model.merger.*, etc. from
                            safetensors shards (header scan only, no model load)
                            Removes ~0.19 GB (0.8B) / ~0.62 GB (2B, 4B) / ~0.85 GB
(9B)
2. Copy side-car files    — tokenizer.json, vocab.json, merges.txt, …
                            Vision-only sidecars (preprocessor_config.json etc.)
omitted
3. Rebuild config.json    — starts from config["text_config"] as root to avoid
                            inheriting vision dimensions from the VLM config.
                            Sets architectures: ["Qwen3_5ForCausalLM"]
                            BNB mode: carries quantization_config, removes
                            model.visual from llm_int8_skip_modules
                            f16 mode:  drops quantization_config entirely
4. Patch chat template    — strips image/video branches from Jinja2 template
                            in both tokenizer_config.json and chat_template.jinja
5. Verify                 — structural check (config + safetensors key scan)
                            + inference check with automatic device selection
```

Device selection (automatic, no flags needed)

Both BNB and f16 modes pick the best available device automatically:

`_pick_device()` result	Threshold	Strategy
`cuda_direct` — full model fits	total ≤ 95% VRAM	CPU load → `_move_to_cuda()`
`cuda_drop` — backbone fits after strip	rest ≤ 95% VRAM	CPU load → `_move_to_cuda()`
`cpu` — too large or no CUDA	rest > 95% VRAM	stay on CPU

`_move_to_cuda()` catches runtime OOM and falls back to CPU — inference always runs regardless. CPU is a valid fallback for both BNB and f16.

HF Hub source

When `--model` is a HF repo id, `snapshot_download` uses the default HF cache (`~/.cache/huggingface/hub`) — re-runs are instant.

---

qwen35-verify / qwen35-verify-qwen35

Both scripts accept a **local path or a HF Hub repo id**. Mode (BNB vs f16) and device strategy are **both auto-detected** — no flags needed.

Source is auto-detected: paths starting with `./`, `../`, `/`, `~` or pointing to an existing directory are treated as local. Anything else containing a `/` is treated as a HF repo id.

`verify_model.py` is the generic base; `verify_qwen35.py` imports it and overrides only what's Qwen3.5-specific:

```
qwen35_toolkit/verify.py     ← generic, uses AutoModelForCausalLM
  │  is_hf_repo()                    detects local path vs HF repo id
  │  resolve_source()                returns (is_remote, label) for display
  │  check_config()                  fetches + validates config.json
  │  model_size_gb()                 actual on-disk size (all weights)
  │  detect_mode()                   reads config.json → "bnb" | "f16"
  │  detect_arch()                   reads config.json → "vlm" | "text"
  │  _pick_device()                  estimates total size vs VRAM → strategy
  │  _move_to_cuda()                 OOM-safe model.cuda() with CPU fallback
  │  restore_visual_to_fp()          restores visual layers to exact bf16
  │  drop_visual_tower()             removes visual submodule, frees RAM/VRAM
  │  count_quantized_layers()        counts Linear4bit vs total nn.Linear
  │  run_inference_tasks()           runs prompt list, reports tok/s
  │  verify()                        full pipeline: steps 1–5
  │
  └── qwen35_toolkit/verify_qwen35.py  ← Qwen3.5 specific, four modes (arch × mode)
        load_qwen35()                 detects arch+mode, picks loader:
                                        vlm+bnb  → Qwen3_5ForConditionalGeneration
```

```
                                        vlm+f16  → Qwen3_5ForConditionalGeneration
                                        text+bnb → AutoModelForCausalLM
                                        text+f16 → AutoModelForCausalLM
          verify_qwen35()             4.  Image inference (VLM only)
                                      5a. Inference THINKING OFF
                                      5b. Inference THINKING ON
```

Verification steps

`verify_model.py` (generic):

```
── 1. Config ──────── BNB: validates quantization_config, checks for expected keys
                      f16: reports architectures / model_type / torch_dtype
── 2. Load ────────── loads model + tokenizer; picks strategy, moves to device
                      _move_to_cuda() used as runtime OOM guard in all paths
── 3. Precision ───── BNB: counts Linear4bit layers, checks visual tower
                      f16: reports linear layer count + weight dtype
── 4. Inference [IMAGE] ── skipped (use verify_qwen35.py for VLM)
── 5. Inference ─────────── model generates a coherent response
                      for each test task
```

`verify_qwen35.py` (Qwen3.5 — all four arch × mode combinations):

```
── 1. Config ──────── BNB: validates quantization_config, checks for expected keys
                      f16: reports architectures / model_type / torch_dtype
── 2. Load ────────── arch + mode auto-detected; loader chosen accordingly:
                        vlm+bnb  → Qwen3_5ForConditionalGeneration +
restore_visual_to_fp
                        vlm+f16  → Qwen3_5ForConditionalGeneration +
torch_dtype=auto
                        text+bnb → AutoModelForCausalLM
                        text+f16 → AutoModelForCausalLM + torch_dtype=auto
── 3. Precision ────── BNB: counts Linear4bit layers, checks visual tower
                      f16: reports linear layer count + weight dtype
── 4. Inference  [IMAGE] ─────────── visual pipeline e2e test (see below)
                      device: GPU for cuda_direct, CPU for cuda_drop / cpu
                      vlm: visual pipeline e2e — device: GPU (cuda_direct) or CPU
                      text: skipped
── 5a. Inference  [THINKING OFF] ── math task, checks response correctness
── 5b. Inference  [THINKING ON]  ── same task with <think> block enabled
```

Automatic device strategy

`_pick_device()` reads actual on-disk file sizes (no model load) and picks a loading strategy. Works identically for BNB and f16 — only the load path differs (f16 loads directly to device; BNB always CPU-first). Size accounting uses the full file size — this captures backbone, visual tower, `lm_head`, `embed_tokens`, and all other non-quantized weights. Summing only the quantized backbone would undercount (e.g. in 9B `lm_head.weight` alone is ~1 GB of fp16).

Threshold	Strategy	Description
total ≤ 95% VRAM	cuda_direct	full model fits → move everything to GPU
(total - visual) ≤ 95% VRAM	cuda_drop	image test on CPU → drop visual → GPU
(total - visual) > 95% VRAM	cpu	stay on CPU throughout

```
cuda_direct:
    Load to CPU (always CPU-first for BNB)
    _move_to_cuda()  — full model to GPU, visual included
    Image test on GPU  ✅ fast
    drop_visual_tower() → free VRAM
    Text inference on GPU

cuda_drop:
    Load to CPU
    Image test on CPU  (visual intact, all on same device)
    drop_visual_tower() → free RAM
    _move_to_cuda()  — backbone + lm_head + embeddings fit in VRAM
    Text inference on GPU

cpu:
    Load to CPU
    Image test on CPU
    drop_visual_tower() → free RAM
    Text inference on CPU  ← BNB does not require CUDA
```

`_move_to_cuda()` **— runtime OOM guard**: wraps every `model.cuda()` call. If `OutOfMemoryError` is raised (e.g. due to fragmented VRAM, driver overhead, or an imprecise file-size estimate), it clears the cache and falls back to CPU. Inference always runs — no step is ever skipped due to VRAM constraints.

Note: `lm_head.weight` is intentionally left at full precision (listed in `llm_int8_skip_modules` for stability / quality — quantizing the output projection degrades generation noticeably). It maps hidden states → vocabulary logits. Visible in all model sizes; ~1 GB in 9B which is why the full-file-size accounting matters there.

Visual tower handling

The converter writes `llm_int8_skip_modules: ["lm_head", "model.visual"]` with exact model-tree paths into `config.json`. Transformers matches by prefix, so `"model.visual"` covers the entire visual tower and prevents re-quantization on load. This is the primary mechanism.

`restore_visual_to_fp()` acts as a **fallback** for models converted before this fix or loaded from older Hub checkpoints. It reloads exact original bf16 weights directly from safetensors on disk (no dequantize rounding error). Step 3 confirms whether restore was needed.

> **Note:** This assumes the remaining text backbone is architecturally compatible with the standalone text config of the same model size. Mixing visual weights from one size with text weights from another is unsupported and will produce garbage output without an obvious error.

Image inference test (step 4)

A 448×224 PNG with the left half red and right half blue is embedded as base64 in the script (no file I/O, no network). The model is asked: `"What color is the left half of this image? Answer with one word."` Expected answer: `"red"` (case-insensitive).

This verifies the full visual pipeline end-to-end: `image tokens → visual encoder → merger → language model → response`

To support a new model family: create `verify_<family>.py`, import helpers from `verify_model`, override only the loader and model-specific generation kwargs.

---

llama.cpp/convert_hf_to_gguf.py

Converts a HuggingFace text-only f16 model to GGUF format. Part of the llama.cpp git submodule — requires the submodule to be initialised. No build needed for this script (Python only).

Input must be a **text-only** model (`Qwen3_5ForCausalLM`) — the VLM variant will fail or produce incorrect output because llama.cpp does not support the visual tower.

```
python llama.cpp/convert_hf_to_gguf.py <model_path> \
    --outtype f16 \
    --outfile <o>.gguf
```

---

llama.cpp/build/bin/llama-quantize

Quantizes a GGUF f16 file to a smaller format. Requires llama.cpp to be built.

Quants

File	Type	Size vs f16	Notes
`Qwen3.5-text-{SIZE}-Q8_0.gguf`	Q8_0	~53% of f16	near-lossless — for high-quality inference
`Qwen3.5-text-{SIZE}-Q6_K.gguf`	Q6_K	~41% of f16	excellent quality, good balance with f16
`Qwen3.5-text-{SIZE}-Q5_K_M.gguf`	Q5_K_M	~37% of f16	very good quality, smaller than Q6
`Qwen3.5-text-{SIZE}-Q4_K_M.gguf`	Q4_K_M	~31% of f16	✅ recommended — best size/quality balance
`Qwen3.5-text-{SIZE}-Q4_K_S.gguf`	Q4_K_S	~30% of f16	optional — slightly smaller, slightly lower quality

```
./llama.cpp/build/bin/llama-quantize <input.gguf> <output.gguf> <TYPE>
```

---

qwen35-inspect

Reads safetensors shard headers (no weights loaded) and prints all weight prefixes with their cumulative sizes. Runs instantly on any model size.

```
qwen35-inspect ./Qwen3.5-4B-bnb-4bit
```

```
Found 5 shard(s)

PREFIX                                                SIZE
──────────────────────────────────────────────────────────
model.language_model                               3.54 GB
model.visual                                       0.89 GB
lm_head                                           90.0 MB
model.visual.merger                               12.3 MB

TOTAL                                              4.54 GB
```

Useful for confirming which components are present before and after `qwen35-strip`.

---

qwen35-render-mermaid

Pre-renders every ```mermaid block in a Markdown file to PNG using the Mermaid CLI (`mmdc`), then outputs a new `.md` file with blocks replaced by `![]()` image references. Needed for PDF export from VS Code Office Viewer (which doesn't render Mermaid natively).

```
# One-time setup
npm install -g @mermaid-js/mermaid-cli
npx puppeteer browsers install chrome-headless-shell

# Render (replace <doc> with the target filename)
qwen35-render-mermaid --input <doc>.md
# → <doc>_pdf.md
# → diagrams/diagram_01.png, diagram_02.png  (mmdc_config.json auto-deleted)

# Higher scale for print quality (default: 2.0)
qwen35-render-mermaid --input <doc>.md --scale 3.0
```

---

qwen35-upload

Six sync modes controlled by `--mode`:

Mode	Direction	When to use	What happens
`init` (default)	push	First push of a new model	Uploads full directory. Hub deduplicates by SHA256 — re-running is safe, unchanged shards not re-transferred
`files`	push	Patching specific files	Single atomic commit of the listed files only. Requires `--files`
`diff`	push	Incremental push after local edits	Compares local SHA256 vs remote, uploads new/changed files, deletes remote-only files
`check`	push dry-run	Inspect before pushing	Same per-file table as `diff`, but nothing is uploaded or deleted
`pull`	pull	Sync local from Hub	Downloads new/changed files, removes local-only files. Large shards skipped if SHA256 already matches
`fetch`	pull dry-run	Inspect before pulling	Same per-file table as `pull`, filesystem untouched

For `.safetensors` / `.gguf` shards SHA256 is read directly from the LFS pointer — no download needed for the push side.

Example `check` / `diff` output (push direction):

```
   FILE                                              SIZE   STATUS
   ──────────────────────────────────────────────────────────────────────────
   config.json                                      2.1 KB  📝 changed
   model-00001-of-00002.safetensors                 4.9 GB  ✅ unchanged
(a3f1c2d4e5b6…)
   model-00002-of-00002.safetensors                 3.1 GB  ✅ unchanged
(9b8e7f6a5c4d…)
   tokenizer_config.json                            1.8 KB  ✅ unchanged
(c2d4e5b6a7f8…)
   README.md                                          512 B  ✨ new
   preprocessor_config.json                             —  🗑️  remote-only → delete

   ── Summary ──────────────────────────────────────────────────────────────
      to upload  : 2
      to delete  : 1
      unchanged  : 3
```

Example `fetch` / `pull` output (pull direction):

```
   FILE                                              SIZE   STATUS
   ──────────────────────────────────────────────────────────────────────────
   config.json                                      2.1 KB  📝 changed
   model-00001-of-00002.safetensors                 4.9 GB  ✅ unchanged
(a3f1c2d4e5b6…)
   model-00002-of-00002.safetensors                 3.1 GB  ✅ unchanged
(9b8e7f6a5c4d…)
   tokenizer_config.json                            1.8 KB  ✅ unchanged
(c2d4e5b6a7f8…)
   some_local_extra.json                                —  🗑️   local-only → remove

   ── Summary ──────────────────────────────────────────────────────────────
      to download : 1
      to remove   : 1
      unchanged   : 3
```

Usage examples

```
# ── Step 1a : BNB 4-bit quantization (keeps visual tower) ──────────────────
qwen35-convert \
    --model  unsloth/Qwen3.5-0.8B \
    --output ./Qwen3.5-0.8B-bnb-4bit

# Low VRAM — quantizes one layer at a time (peak VRAM ~400 MB)
qwen35-convert \
    --model    unsloth/Qwen3.5-9B \
    --output   ./Qwen3.5-9B-bnb-4bit \
    --low-vram

# Verbose — print every layer replacement and skip decision
qwen35-convert \
```

```
    --model   unsloth/Qwen3.5-0.8B \
    --output  ./Qwen3.5-0.8B-bnb-4bit \
    --verbose

# ── Step 1b : strip visual tower ──────────────────────────────────────
# f16 VLM → text-only bf16
qwen35-strip \
    --model  unsloth/Qwen3.5-0.8B \
    --output ./Qwen3.5-text-0.8B \
    --mode   f16

# BNB 4-bit VLM → text-only BNB 4-bit (from already-converted local model)
qwen35-strip \
    --model  ./Qwen3.5-0.8B-bnb-4bit \
    --output ./Qwen3.5-text-0.8B-bnb-4bit \
    --mode   bnb

# From HF Hub with token
qwen35-strip \
    --model    unsloth/Qwen3.5-9B \
    --output   ./Qwen3.5-text-9B \
    --mode     f16 \
    --hf-token hf_...

# ── Inspect safetensors contents (optional, before/after strip) ──────
qwen35-inspect ./Qwen3.5-0.8B-bnb-4bit
qwen35-inspect ./Qwen3.5-text-0.8B-bnb-4bit

# ── Step 2 : verify ───────────────────────────────────────────────────
# Qwen3.5-specific: thinking ON/OFF + image inference test
qwen35-verify-qwen35 --model ./Qwen3.5-0.8B-bnb-4bit
qwen35-verify        --model ./Qwen3.5-text-0.8B-bnb-4bit

# From HF Hub (cached — re-runs skip download)
qwen35-verify-qwen35 --model techwithsergiu/Qwen3.5-0.8B-bnb-4bit
qwen35-verify        --model techwithsergiu/Qwen3.5-text-0.8B-bnb-4bit

# 9B model on 7.7 GB card — auto-detects cuda_drop: image test on CPU,
# drops visual tower, then backbone + lm_head moved to GPU
qwen35-verify-qwen35 --model ./Qwen3.5-9B-bnb-4bit

# ── Step 3 : GGUF conversion ──────────────────────────────────────────
# text-only f16 → GGUF f16 (requires llama.cpp — see Setup)
python llama.cpp/convert_hf_to_gguf.py ./Qwen3.5-text-0.8B \
    --outtype f16 \
    --outfile ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-F16.gguf

# Quantize — Q4_K_M main, Q4_K_S optional
# Q8_0 / Q6_K / Q5_K_M optional
./llama.cpp/build/bin/llama-quantize \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-F16.gguf \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-Q8_0.gguf \
    Q8_0

./llama.cpp/build/bin/llama-quantize \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-F16.gguf \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-Q6_K.gguf \
    Q6_K

./llama.cpp/build/bin/llama-quantize \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-F16.gguf \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-Q5_K_M.gguf \
```

```
    Q5_K_M

./llama.cpp/build/bin/llama-quantize \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-F16.gguf \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-Q4_K_M.gguf \
    Q4_K_M

./llama.cpp/build/bin/llama-quantize \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-F16.gguf \
    ./Qwen3.5-text-0.8B-GGUF/Qwen3.5-text-0.8B-Q4_K_S.gguf \
    Q4_K_S

# ── PDF export (optional) ──────────────────────────────────────────
qwen35-render-mermaid --input qwen35_conversion_pipeline.md
# → qwen35_conversion_pipeline_pdf.md + diagrams/

# ── Step 4 : upload to HF Hub ──────────────────────────────────────
# BNB / f16 models — first push
qwen35-upload \
    --local ./Qwen3.5-0.8B-bnb-4bit \
    --repo  techwithsergiu/Qwen3.5-0.8B-bnb-4bit

# GGUF quants — first push (all quants in one repo)
qwen35-upload \
    --local ./Qwen3.5-text-0.8B-GGUF \
    --repo  techwithsergiu/Qwen3.5-text-0.8B-GGUF

# First push — private repo
qwen35-upload \
    --local    ./Qwen3.5-0.8B-bnb-4bit \
    --repo     techwithsergiu/Qwen3.5-0.8B-bnb-4bit \
    --private \
    --hf-token hf_...

# Patch specific files only (no re-upload of shards)
qwen35-upload \
    --local   ./Qwen3.5-0.8B-bnb-4bit \
    --repo    techwithsergiu/Qwen3.5-0.8B-bnb-4bit \
    --mode    files \
    --files   README.md tokenizer_config.json

# Push only new/changed files (+ delete remote-only files)
qwen35-upload \
    --local ./Qwen3.5-0.8B-bnb-4bit \
    --repo  techwithsergiu/Qwen3.5-0.8B-bnb-4bit \
    --mode  diff

# Dry-run push — inspect delta without uploading
qwen35-upload \
    --local ./Qwen3.5-0.8B-bnb-4bit \
    --repo  techwithsergiu/Qwen3.5-0.8B-bnb-4bit \
    --mode  check

# Pull new/changed files from Hub
qwen35-upload \
    --local ./Qwen3.5-0.8B-bnb-4bit \
    --repo  techwithsergiu/Qwen3.5-0.8B-bnb-4bit \
    --mode  pull

# Dry-run pull
qwen35-upload \
    --local ./Qwen3.5-0.8B-bnb-4bit \
```

```
    --repo  techwithsergiu/Qwen3.5-0.8B-bnb-4bit \
    --mode  fetch
```

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